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SOLIDWORKS
Sustainability Report

[company name here] [city, state here] [company url here]

[name] · [title] · [email address] · (###) ### ####



Model Name: Assembly Drone

Weight: 1255.43 g

Built to last: 1.0 year

Duration of use: 1.0 year



Manufacturing Region
The choice of manufacturing region determines the energy sources and technologies used in the modeled material creation and manufacturing steps of the product's life cycle.

Use Region
The use region is used to determine the energy sources consumed during the product's use phase (if applicable) and the destination for the product at its end-of-life. Together with the manufacturing region, the use region is also used to estimate the environmental impacts associated with transporting the product from its manufacturing location to its use location.

Summary

[Abstract]

Learn more about Life Cycle Assessment

Sustainability Report

Model Name:	Assembly Drone	Weight:	1255.43 g
		Built to last:	1.0 year
		Duration of use:	1.0 year

Assembly Process

Region:	Asia
Energy type:	None
Energy amount:	0.00 kWh
Built to last:	1.0 year

Use

Region:	Asia
Energy type:	None
Energy amount:	0.00 kWh
Duration of use:	1.0 year

Transportation

Truck distance:	1600 km
Train distance:	0.00 km
Ship distance:	0.00 km
Airplane Distance:	0.00 km

End of Life

Recycled:	33 %
Incinerated:	13 %
Landfill:	54 %

Comments

Click here for alternative units such as 'Miles Driven in a Car'

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Environmental Impact (calculated using CML impact assessment methodology)

Carbon Footprint



570 kg CO_2e

Material:	18 kg CO_2e
Manufacturing:	550 kg CO_2e
Use:	0.00 kg CO_2e
Transportation:	0.107 kg CO_2e
End of Life:	0.690 kg CO_2e

Total Energy Consumed



5800 MJ

Material:	260 MJ
Manufacturing:	5500 MJ
Use:	0.00 MJ
Transportation:	1.4 MJ
End of Life:	0.507 MJ

Air Acidification



7.9 kg SO_2e

Material:	0.092 kg SO_2e
Manufacturing:	7.8 kg SO_2e
Use:	0.00 kg SO_2e
Transportation:	5.4E-4 kg SO_2e
End of Life:	3.5E-4 kg SO_2e

Water Eutrophication



0.308 kg PO_4e

Material:	5.2E-3 kg PO_4e
Manufacturing:	0.302 kg PO_4e
Use:	0.00 kg PO_4e
Transportation:	1.1E-4 kg PO_4e
End of Life:	8.7E-4 kg PO_4e

Material Financial Impact

2.20 USD

Comments

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Component Environmental Impact

Top Ten Components Contributing Most to the Four Areas of Environmental Impact

Component	Carbon	Water	Air	Energy
Part4 Companion Computer V2	220	0.116	3.0	2200
Part19 Camera Sphere	72	0.039	1.0	720
pixhawk 4 (V)	49	0.026	0.680	490
structure core frame (V)	43	0.023	0.593	430
Part8 Frame T265	16	8.5E-3	0.220	160
Part18 Lidar	14	7.4E-3	0.193	140
servo motor F60PRO IV V2.0 KV1750 (V)	10	5.4E-3	0.140	100
Camera Arm (V)	7.9	4.2E-3	0.110	79
Part20 Camera Base Arm	6.0	3.2E-3	0.084	60
Part17 Lidar Mount	5.4	2.9E-3	0.075	54



Comments

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RAW MATERIAL
EXTRACTION
MATERIAL
PROCESSING
PART
MANUFACTURING
ASSEMBLY
PRODUCT
USE
END OF LIFE
SUSTAINABLE
DESIGN

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Glossary

Air Acidification - Sulfur dioxide, nitrous oxides other acidic emissions to air cause an increase in the acidity of rainwater, which in turn acidifies lakes and soil. These acids can make the land and water toxic for plants and aquatic life. Acid rain can also slowly dissolve manmade building materials such as concrete. This impact is typically measured in units of either kg **sulfur dioxide equivalent (SO_2), or moles H+ equivalent**.

Carbon Footprint - Carbon-dioxide and other gasses which result from the burning of fossil fuels accumulate in the atmosphere which in turn increases the earth's average temperature. Carbon footprint acts as a proxy for the larger impact factor referred to as Global Warming Potential (GWP). Global warming is blamed for problems like loss of glaciers, extinction of species, and more extreme weather, among others.

Total Energy Consumed - A measure of the non-renewable energy sources associated with the part's lifecycle in units of megajoules (**MJ**). This impact includes not only the electricity or fuels used during the product's lifecycle, but also the upstream energy required to obtain and process these fuels, and the embodied energy of materials which would be released if burned. PED is expressed as the net calorific value of energy demand from non-renewable resources (e.g. petroleum, natural gas, etc.). Efficiencies in energy conversion (e.g. power, heat, steam, etc.) are taken into account.

Water Eutrophication - When an over abundance of nutrients are added to a water ecosystem, eutrophication occurs. Nitrogen and phosphorous from waste water and agricultural fertilizers causes an overabundance of algae to bloom, which then depletes the water of oxygen and results in the death of both plant and animal life. This impact is typically measured in either kg **phosphate equivalent (PO_4) or kg nitrogen (N) equivalent**.

Life Cycle Assessment (LCA) - This is a method to quantitatively assess the environmental impact of a product throughout its entire lifecycle, from the procurement of the raw materials, through the production, distribution, use, disposal and recycling of that product.

Material Financial Impact - This is the financial impact associated with the material only. The mass of the model is multiplied by the financial impact unit (units of currency/units of mass) to calculate the financial impact (in units of currency).

Learn more about Life Cycle Assessment